AE 121A Set 4

3.05 :

a) $\dfrac{\dot m}{A^*} = \dfrac{P_0\,\gamma^{1/2}}{(R T_0)^{1/2}} \left(\dfrac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}}$ $R = 8.314\ \frac{J}{mol\,K}$ when $M = 1$ so $A^* = A_t$

Viking 5C:

expansion ratio : $\frac{A_e}{A_t} = 10.5$, $\dot m : 275.2\ kg/s$, $T_0 = 3350\,K$, $P_0 : 5800\,kPa$, $\gamma = 1.2$

Solved in part b

$\dfrac{275.2\ kg/s}{A^*} = \dfrac{5800\,kPa \cdot 1.2^{1/2}}{(361.48\ \frac{J}{kg\,K} \cdot 3350\,K)^{1/2}} \left(\dfrac{2}{1.2+1}\right)^{\frac{1.2+1}{2(1.2-1)}}$ $\dfrac{A_e}{0.0805\,m^2} = 10.5$

$\dfrac{275.2\ kg/s}{A^*} = 5773.7 \cdot 0.592025$ $A_e = 0.84\,m^2$

$A^* = 0.0805\,m^2$

$\boxed{A_e = 0.84\ m^2 \quad A_t = 0.0805\ m^2}$

Viking 4B: expansion ratio of $30.8 = \frac{A_e}{A_t}$, mass flow rate of $278.0\ kg/s$ $P_0 = 5850\,kPa$

$\dfrac{278.0\ kg/s}{A^*} = \dfrac{5850\,kPa \cdot 1.2^{1/2}}{(361.48\ \frac{J}{kg\,K} \cdot 3350\,K)^{1/2}} \left(\dfrac{2}{1.2+1}\right)^{\frac{1.2+1}{2(1.2-1)}}$ $\dfrac{A_e}{0.0798\,m^2} = 30.8$

$\dfrac{278.0\ kg/s}{A^*} = 5823.5 \cdot 0.592025$ $A_e = 2.45\,m^2$

$A^* = 0.0798\,m^2$

$\boxed{A_e = 2.45\ m^2 \quad A_t = 0.0798\ m^2}$

b) **Viking 5C:**

$\dfrac{A_e}{A_t} = 10.5 = \dfrac{1}{M}\left[\dfrac{2}{1.2+1}\left(1 + \dfrac{1.2-1}{2} M^2\right)\right]^{\frac{1.2+1}{2(1.2-1)}} \Rightarrow 10.5 = \dfrac{1}{M}\left(0.909\,(1 + 0.1M^2)\right)^{5.5} \Rightarrow$ Wolfram alpha $\Rightarrow$

$\boxed{M = 0.0564824}$ or $\boxed{M = 3.31236}$

$\dfrac{T_e}{T_{tot}} = \dfrac{T_e}{T_0} = \dfrac{T_e}{3350\,K} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-1} = \left(1 + \frac{1.2-1}{2}(3.31236)^2\right)^{-1} = (2.0972)^{-1} = 0.4768$

$\dfrac{P_e}{P_{tot}} = \dfrac{P_e}{P_0} = \dfrac{P_e}{5800\,kPa} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{\gamma}{\gamma-1}} = (2.0972)^{-\frac{1.2}{1.2-1}} = 0.011753329$

$\dfrac{\rho_e}{\rho_{tot}} = \dfrac{\rho_e}{\rho_0} = \dfrac{\rho_e}{4.79\ \frac{kg}{m^3}} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{1}{\gamma-1}} = (2.0972)^{-\frac{1}{1.2-1}} = 0.024649082$

$\rho_0 = \dfrac{P_0}{R T_0}$ for an isentropic process

$R = 8.314\ \frac{J}{mol\,K} \cdot \dfrac{1}{0.023\ \frac{kg}{mol}}$

$R = 361.48\ \frac{J}{kg\,K}$

$\rho_0 = \dfrac{5800 \times 10^3\ Pa}{361.48\ \frac{J}{kg\,K} \cdot 3350\,K} = 4.79\ \frac{kg}{m^3}$

$\boxed{T_e = 1597.39\,K \quad P_e = 68.17\,kPa \quad \rho_e = 0.1180597\ \frac{kg}{m^3}}$ ▷ Supersonic

$\dfrac{T_e}{T_{tot}} = \dfrac{T_e}{T_0} = \dfrac{T_e}{3350\,K} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-1} = \left(1 + \frac{1.2-1}{2}(0.05648)^2\right)^{-1} = 0.9996810756$

$\dfrac{P_e}{P_{tot}} = \dfrac{P_e}{P_0} = \dfrac{P_e}{5800\,kPa} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{\gamma}{\gamma-1}} = (1.000319026)^{-\frac{1.2}{1.2-1}} = 0.9980879795$

$\dfrac{\rho_e}{\rho_{tot}} = \dfrac{\rho_e}{\rho_0} = \dfrac{\rho_e}{4.79\ \frac{kg}{m^3}} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{1}{\gamma-1}} = (1.000319026)^{-\frac{1}{1.2-1}} = 0.9984063955$

$\boxed{T_e = 3348.93\,K \quad P_e = 5788.9\,kPa \quad \rho_e = 4.7824\ \frac{kg}{m^3}}$ ▷ Subsonic

Viking 4B :

$$\frac{A_e}{A_t} = 30.8 = \frac{1}{M}\left[\frac{2}{1.2+1}\left(1+\frac{1.2-1}{2}M^2\right)\right]^{\frac{1.2+1}{2(1.2-1)}} \Rightarrow 30.8 = \frac{1}{M}\left(0.909\left(1+0.1M^2\right)\right)^{5.5} \Rightarrow \text{Wolfram alpha} \Rightarrow$$

$$\boxed{M = 0.0192255} \quad \text{or} \quad \boxed{M = 4.05738}$$

$$\frac{T_e}{T_{tot}} = \frac{T_e}{T_0} = \frac{T_e}{3350\text{K}} = \left(1+\frac{\gamma-1}{2}M_e^2\right)^{-1} = \left(1+\frac{1.2-1}{2}(4.05738)^2\right)^{-1} = (2.64623)^{-1} = 0.3779$$

$$\frac{P_e}{P_{tot}} = \frac{P_e}{P_0} = \frac{P_e}{5850\text{kPa}} = \left(1+\frac{\gamma-1}{2}M_e^2\right)^{-\frac{\gamma}{\gamma-1}} = (2.64623)^{-\frac{1.2}{1.2-1}} = 0.002912$$

$$\frac{\rho_e}{\rho_{tot}} = \frac{\rho_e}{\rho_0} = \frac{\rho_e}{4.83\frac{\text{kg}}{\text{m}^3}} = \left(1+\frac{\gamma-1}{2}M_e^2\right)^{-\frac{1}{\gamma-1}} = (2.64623)^{-\frac{1}{1.2-1}} = 0.007707$$

$\rho_0 = \frac{P_0}{RT_0}$ for an isentropic process

$$\rho_0 = \frac{5850\times10^3\ \text{Pa}}{361.46\frac{\text{J}}{\text{mol K}}\cdot 3350\ \text{K}} = 4.831\ \frac{\text{kg}}{\text{m}^3}$$

$$\boxed{T_e = 1265.95\ \text{K} \qquad P_e = 17.036\ \text{kPa} \qquad \rho_e = 0.03723\ \frac{\text{kg}}{\text{m}^3}}$$

supersonic

$$\frac{T_e}{T_{tot}} = \frac{T_e}{T_0} = \frac{T_e}{3350\text{K}} = \left(1+\frac{\gamma-1}{2}M_e^2\right)^{-1} = \left(1+\frac{1.2-1}{2}(0.0192255)^2\right)^{-1} = (1.000036962)^{-1} = 0.9999630394$$

$$\frac{P_e}{P_{tot}} = \frac{P_e}{P_0} = \frac{P_e}{5850\text{kPa}} = \left(1+\frac{\gamma-1}{2}M_e^2\right)^{-\frac{\gamma}{\gamma-1}} = (1.000036962)^{-\frac{1.2}{1.2-1}} = 0.9997782567$$

$$\frac{\rho_e}{\rho_{tot}} = \frac{\rho_e}{\rho_0} = \frac{\rho_e}{4.83\frac{\text{kg}}{\text{m}^3}} = \left(1+\frac{\gamma-1}{2}M_e^2\right)^{-\frac{1}{\gamma-1}} = (1.000036962)^{-\frac{1}{1.2-1}} = 0.9998152105$$

$$\boxed{T_e = 3349.88\ \text{K} \qquad P_e = 5848.70\ \text{kPa} \qquad \rho_e = 4.829107\ \frac{\text{kg}}{\text{m}^3}}$$

subsonic